THE SHEAF OF HOMOMORPHISMS FROM THE PUSHFORWARD OF THE STRUCTURE SHEAF ALONG AN AFFINE MORPHISM

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0BUZ (sources/stacks-duality/pinned-original-source.tex line 2705).

Allowed standard prerequisites (editorial).

- Adjunction between tensor product and internal hom of sheaves of modules. Modules, Lemma labelled modules-lemma-adjoint-tensor-restrict, cited at proof lines 2717-2719. Gives the adjunction.
- Comparison of quasi-coherent sheaves on an affine scheme with modules. Schemes, Lemma labelled schemes-lemma-compare-constructions, cited at proof lines 2719-2724. Gives the description of the sections over an affine open.

Source reference. Original lemma 0A6F: sources/stacks-chow/tags.txt, source lines 12843–12843. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 2719 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 01I7: sources/stacks-chow/tags.txt, source lines 1968–1968. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 2723 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *With notation as above. The functor $\mathcal{H}om(f_*\mathcal{O}_Y, -)$ is a right adjoint to the restriction functor $Mod(f_*\mathcal{O}_Y) \to Mod(\mathcal{O}_X)$. For an affine open $U \subset X$ we have*

$$\Gamma(U, \mathcal{H}om(f_*\mathcal{O}_Y, \mathcal{F})) = \mathrm{Hom}_A(B, \mathcal{F}(U))$$

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where $A = \mathcal{O}_X(U)$ *and* $B = \mathcal{O}_Y(f^{-1}(U))$.

Proof. Adjointness follows from Modules, Lemma 0A6F. As f is affine we see that $f_*\mathcal{O}_Y$ is the quasi-coherent sheaf corresponding to B viewed as an A-module. Hence the description of sections over U follows from Schemes, Lemma 01I7. □